1. What is Artificial Intelligence (AI)?

Answer: AI is a technology that enables computers to perform tasks that normally require human intelligence, such as learning, reasoning, and decision-making.

2. What is Machine Learning (ML)?

Answer: Machine Learning is a part of AI where computers learn patterns from data and use them to make predictions or decisions without being explicitly programmed for every task.

3. What are the main types of Machine Learning?

Answer: The three main types are:

Supervised Learning

Unsupervised Learning

Reinforcement Learning

4. What is Supervised Learning?

Answer: Supervised learning uses labeled data to train a model. For example, images labeled as "ripe" and "unripe" can be used to train a classification model.

5. What is Unsupervised Learning?

Answer: Unsupervised learning works with data that does not have predefined labels. The model finds patterns or groups in the data.

6. What is a dataset?

Answer: A dataset is a collection of data used to train, test, or evaluate a machine-learning model.

7. What is classification?

Answer: Classification is an ML technique that places data into predefined categories. For example, classifying a fruit as ripe, unripe, or overripe.

8. What is regression?

Answer: Regression is used to predict a numerical value, such as predicting temperature, price, or sales.

9. What is a neural network?

Answer: A neural network is a machine-learning model inspired by the human brain. It consists of connected nodes called neurons that learn patterns from data.

10. What is overfitting?

Answer: Overfitting happens when a model learns the training data too closely and performs poorly on new, unseen data.